

De Montfort University
Beng (Hons) Mechanical Engineering – Year 3
Module: Advanced Mechanical Materials and Professional Practice - ENGE3001
Experiment: Fatigue Failure and Testing of Materials

Lab Report: Effects of Fatigue on Different Materials

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Course: Mechanical Engineering
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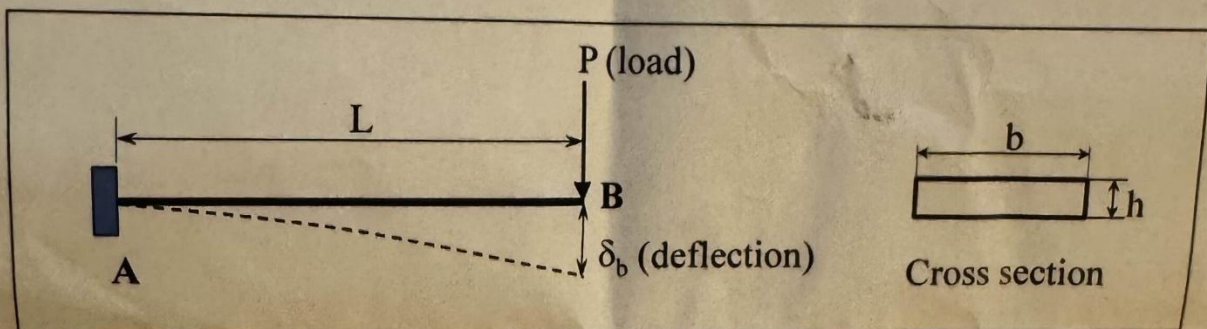
ENGE3001: Advanced Mechanical Materials and Professional Practice

Fatigue Assignment: Fatigue Failure and Testing of Material

LOG SHEET

Name: *Feras*

Date and time: *27/0ct/2025*



1. At the beginning of the experiment, please measure the following:

Steel:
 $L = 40 \text{ mm}$ *96-92 cm*
 $b = 10.22$
 $h = 0.92$
 $\delta_b = 18/2 = 9 \text{ mm}$

Aluminium:
 40 mm
 10 mm
 1.09 mm
 9 mm

CFRP:
 40 mm
 10.01 mm
 1.03 mm
 9 mm

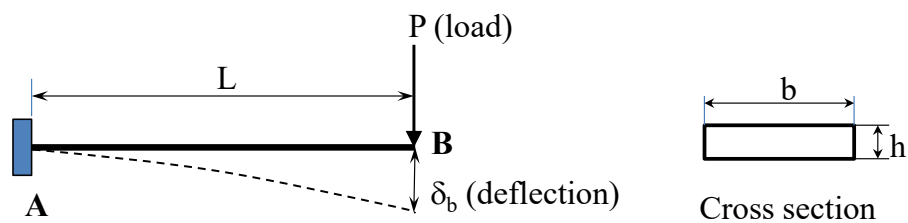
2. Now conduct the fatigue tests on the steel sample, aluminium sample, and CFRP sample and record the number of cycles to failure for each sample:

Samples	Cycles to failure (Test 1)	Cycles to failure (Test 2)
Steel	<i>3237</i>	<i>3294</i>
Aluminium	<i>386</i>	<i>253</i>
CFRP	<i>125,000</i>	<i>125,000</i>

ENGE3001: Advanced Mechanical Materials and Professional Practice
Fatigue Failure and Testing of Material
LAB SHEET

Name and Student ID: **P2731284, Feras Bouteraa**

Date and time: **27/10/2025**



1. At the beginning of the experiment, please measure the following:

Tab. 1. Measurements of subjects by material

	Steel	Aluminium	CFRP
L (mm)	40	40	40
b (mm)	10.22	10	10.01
h (mm)	0.92	1.09	1.03
δ_b (mm)	9	9	9

2. Now conduct the fatigue tests on the steel sample, aluminium sample, and CFRP sample and record the number of cycles to failure for each sample:

Tab. 2. Effect of material on cycles to failure

	Cycles to failure (Test 1)	Cycles to failure (Test 2)	Cycles to failure (average)
Steel	3,237	3,294	3,265
Aluminium	386	253	319.5
CFRP	125,000	125,000	125,000

Abstract – Fatigue is the failure of a material through cyclical loading by progressive and localised damage. This is the number one cause of failure in mechanical systems making it a critical component for success in many engineering industries. In this experiment steel, aluminium and CFRP were put under cyclical loading of uniform deflection. aluminium experienced this fatigue test with the lowest average cycles (320) followed by steel and finally CFRP. CFRP experienced a maximum stress of only (582 MPa). Steel did relatively well compared to aluminium yet experienced much higher stress. This indicates that the properties of steel stabilised the microplastic deformation whereas, aluminium’s microplastic deformation continued to progress. Contrastingly CFRP showed no effects of microplastic deformation, demonstrating superiority in fatigue resistance. In this experiment CFRP withstood 125,000 cycles without breaking compared to steel fracturing at 3,265. However, real world performance can vary based on scenario. To conclude, fatigue resistance is directly connected to the microplastic properties of a material.

Terminology – “CFRP” Carbon Fiber Reinforced Polymer. Microplastic Deformation.

I. Introduction

Material failure is an important topic in every industry. Fatigue is the leading type of failure in mechanical systems that happens due to cyclical or fluctuating stress causing catastrophic failure, often without warning. Fatigue happens when microscopic cracks form and progress, specifically weakening a localised area. It is an important topic for industries such as automotive, aerospace, biomedical among many more. An example of catastrophic fatigue failure is the 1994 San Marino Grand Prix accident involving Formula 1 driver *Ayrton Senna*. A welded inner tube under repeated stress led to a sudden fracture and loss of control of the car. This incident highlights how fatigue in a critical component can become fatal. Fatigue failure typically occurs through three stages: crack initiation, progressive propagation and finally sudden fracture. In this experiment, steel, aluminium and CFRP will be tested under cyclical bending. The cycles to failure of each material will be noted alongside their dimensions before failure. Using the dimensions, deflection amount, cycles to failure and the material’s young’s modulus; we can find important values that can aid in the explanation of what makes a material effective against fatigue.

II. Methodology

Measurements - Callipers were calibrated by setting the displayed measurement to zero when at the minimum position to ensure precision. The subjects were then measured using the callipers alongside the measurement of deflection to calculate second moment of area (I), Load (P), maximum bending moment (M_{max}) and maximum stress (σ_{max}).

Experimental Procedure - The subjects were then secured in the cyclical bending moment machine; one side was lined up to ensure uniform length to a marker at (40mm). the (cycle) counter was then set to zero, the cover was placed for personal safety, and the machine was started. Force was applied parallel to height, and the materials were manipulated well within their elastic region. Once the material reached failure, the machine automatically stopped, and the number of cycles were noted. The materials were measured only the first time however the following procedure was repeated for a total of two tests of each material.

III. Results & Calculations

Note: SI units throughout. Youngs modulus for steel is 205 GPa, aluminium is 72 GPa, and for CFRP is 67 GPa.

$$(1) I = \frac{bh^3}{12}, \text{ e.g. } I_{steel} = \frac{10.22 \times 10^{-3} (9.2 \times 10^{-3})^3}{12} = 6.63 \times 10^{-13} m^4,$$

where (I) represents the second moment of area, (b) is base length and (h) are the height of the subject.

(2) $P = \frac{3EI\delta_b}{L^3}$, e.g. $P_{steel} = \frac{3 \times 205 \times 10^9 \times 9 \times 10^{-3}}{(40 \times 10^{-3})^3} = 57.3 N$, where (P) is load, (L) is the length of the subject, (δ_b) is the deflection & (E) is young’s modulus.

(3) $M_{max} = PL$, e.g. $M_{max}^{steel} = 57.3 \times 40 \times 10^{-3} = 2.29 Nm$, where (M_{max}) is the maximum bending moment.

(4) $\sigma_{max} = \frac{M_{max} y_{max}}{I} = \frac{M_{max} \times \frac{h}{2}}{I}$, $\sigma_{max}^{steel} = \frac{2.29 \times \frac{9.2 \times 10^{-3}}{2}}{6.63 \times 10^{-13}} = 1.59 \times 10^9 Nm^{-2}$, where (σ_{max}) is the maximum stress and (y_{max}) is the perpendicular distance from the neutral axis.

Tab. 3. Calculated values for different materials tested.

	Steel	Aluminium	CFRP
$I (m^4)$	6.63×10^{-13}	1.08×10^{-12}	9.12×10^{-13}
$P (N)$	57.3	32.8	25.8
$M_{max} (Nm)$	2.29	1.31	1.03
$\sigma_{max} (MPa)$	1590	661	582

IV. Discussion

Steel, Aluminium and CFRP are calculated to have a maximum stress shown in (Tab. 3) of 1590 MPa, 661 MPa and 582 MPa respectively. Furthermore, average cycle counts shown in (Tab. 2) of 3237, 386 and 125000. Steel had relatively high stress and cycle count. Aluminium had low stress and low cycle count. CFRP has low stress but a significantly higher cycle count than both. Steel demonstrated more cycles than aluminium (8.4×) and a higher maximum stress (2.4×). Since deflection was kept constant, this must be a demonstration of steel’s relatively high stiffness and a difference between the materials in their microplasticity. Contrastingly, CFRP withstood relatively low bending pressure with exponentially higher cycles. This displays CFRP’s properties provide excellent resistance to crack propagation under cyclical loading especially considering its’ low Young’s Modulus compared to steel. This is in line with Aluminium’s lack of an elastic limit [1], steels elastic limit [2] and CFRP’s linear elastic behaviour [3]. It is important to realise that the localised failure occurred at the area of maximum stress, since the dimensions of the material were uniform relative to the type of stress applied in this experiment. The steel sample failed with no visible necking, smooth area and rough area. Aluminium also failed with no visible necking, or beach marks and much smoother overall than steel.

V. Conclusion

In this experiment, CFRP is clearly superior. Aluminium, lacking a fatigue limit, fails at low cycle counts even under relatively moderate stress, due to its progressive microplastic damage. Steel had a higher cycle count due to its fatigue limit. The exponentially higher cycle count of CFRP is due to the linear elastic behaviour it exhibits.

Highlighting the importance of a materials microplastic properties.

To conclude, fatigue can cause catastrophic failure at relatively low loads and low cycles. It is therefore critical to ensure material selection and component design accounts for fatigue, as catastrophic failure can occur even under moderate cyclic loading.

VI. References

[1] G. T. Yahr, Oak Ridge, Tennessee, date N/A, “Fatigue Design Curves and Analysis for Aluminium”, [online]. Available: https://www.engineersedge.com/materials/fatigue_design_curves__16022.htm

[2] T. Aldeeb, M. Abduelmula, 18 Oct 2018, “Fatigue Strength of S275 Mild Steel under Cyclic Loading”, [online]. Available: https://www.researchgate.net/publication/328355534_Fatigue-Strength-of-S275-Mild-Steel-under-Cyclic-Loading

[3] W. Mu, J. Na, W. Tan, G. Wang, H. Shen, X. Li, Jilin University. Changchun. China, 19 March 2020, “Durability of adhesively bonded CFRP-aluminium alloy joints subjected to coupled temperature and alternating load”, [online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0143749620300440>